Unit: 6 Unit 5: Linear Equations
Topic: Perpendicular Lines

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Q2. If $y = 3x - 2$ and $y = mx + b$ are perpendicular, what is m?
(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 3
(D) -3
(E) 0

Q3. Which line is perpendicular to $y = x + 2$?
(A) $y = x - 2$
(B) $y = -x + 2$
(C) $y = -x - 2$
(D) $y = 2x + 1$
(E) $y = -2x - 1$

Q4. If two lines have slopes m_1 and m_2, and $m_1 \cdot m_2 = -1$, what can be concluded?
(A) The lines are parallel.
(B) The lines are perpendicular.
(C) The lines are vertical.
(D) The lines are horizontal.
(E) The lines intersect.

Q5. What is the equation of a line perpendicular to $y = -4x+1$ and passing through the point $(0, 0)$?
(A) $y = \frac{1}{4}x$
(B) $y = -\frac{1}{4}x$
(C) $y = 4x$
(D) $y = -4x$
(E) $y = 0$

Q6. If $y = 2x - 3$ and $y = mx + b$ are perpendicular, and the lines intersect at $(2, 1)$, what is b?
(A) $-\frac{5}{2}$
(B) $-\frac{3}{2}$
(C) -2
(D) -1
(E) 0

Q7. Which of the following lines is perpendicular to $y = -\frac{3}{4}x + 2$?
(A) $y = \frac{4}{3}x - 1$
(B) $y = -\frac{4}{3}x + 1$
(C) $y = \frac{3}{4}x + 2$
(D) $y = -\frac{3}{4}x - 2$
(E) $y = 0$

Q8. What is the slope of a line that is perpendicular to $y = -2x + 1$ and $y = 3x - 2$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) $-\frac{1}{3}$
(D) $\frac{1}{3}$
(E) 0

Open Ended Questions (FRQ)

Q9. Find the equation of a line that is perpendicular to $y = x - 2$ and passes through the point $(1, 3)$.

Q10. If $y = 2x + 1$ and $y = mx + b$ are perpendicular, and the lines intersect at $(1, 3)$, find the values of m and b.

Chef's Tips

- Tip 1: Ensure students understand the concept of negative reciprocal slopes.
- Tip 2: Emphasize the importance of identifying the relationship between the slopes of perpendicular lines.
- Tip 3: Have students practice writing equations of perpendicular lines using given slopes and points.